UTS UNIVERSITY OF TECHNOLOGY SYDNEY

Faculty of Engineering and IT

Assignment Cover Sheet

Subject: 41277 Control Design

Assignment Number: Demonstration Practical 1 **Date Submitted:** 28/08/2026

Assignment Title: System Modelling

Student Name(s) and Number(s) **Tutorial Group:** 02

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Statement of Collaboration:

This assignment was completed collaboratively by Jaylen Avtarovski, Elina Kim, and Rhyse Williams, with all three members contributing equally to the derivations, analysis, and written responses across Q1, Q2, and Q3. All group discussions, calculations, and write-ups were conducted jointly, and any external tools or resources used have been acknowledged in accordance with UTS academic integrity guidelines.

Contribution Weighting

Student Number	Student Name	Q1	Q2	Q3
24767939	Jaylen Avtarovski	100	100	100
24817532	Rhyse Williams	100	100	100
24949885	Elina Kim	100	100	100

Fill in each team member's percentage contribution to each question before submission.

Answers to the Assignment Questions:

Q1: Cart Pendulum Modelling (20 marks)

Using the required generalised coordinates,

$$q_1 = x_c \qquad\qquad q_2 = \phi$$

where x_c is the horizontal displacement of the cart and ϕ is measured anticlockwise from the negative x-axis.

From the geometry of the system, the pendulum mass position is

$$x_p = x_c - \ell\cos\phi$$
$$y_p = -\ell\sin\phi$$

where ℓ is the distance from the pivot to the centre of oscillation.

Differentiating with respect to time,

$$\dot{x}_p = \dot{x}_c + \ell\dot{\phi}\sin\phi$$
$$\dot{y}_p = -\ell\dot{\phi}\cos\phi$$

The total co-kinetic energy of the system is the sum of the cart and pendulum kinetic energies:

$$T^* = T_c + T_p$$

with

$$T_c = \tfrac{1}{2}M\dot{x}_c^2$$

and

$$T_p = \tfrac{1}{2}m\left(\dot{x}_p^2 + \dot{y}_p^2\right)$$

Therefore,

$$T^* = \frac{1}{2}M\dot{x}_c^2 + \frac{1}{2}m\left[\left(\dot{x}_c + \ell\dot{\phi}\sin\phi\right)^2 + \left(-\ell\dot{\phi}\cos\phi\right)^2\right]$$

Simplifying,

$$T^* = \frac{1}{2}M\dot{x}_c^2 + \frac{1}{2}m\left(\dot{x}_c^2 + 2\ell\dot{x}_c\dot{\phi}\sin\phi + \ell^2\dot{\phi}^2\right)$$
$$T^* = \frac{1}{2}\left(M+m\right)\dot{x}_c^2 + m\ell\dot{x}_c\dot{\phi}\sin\phi + \frac{1}{2}m\ell^2\dot{\phi}^2$$

Taking the pivot height as the gravitational reference,

$$V = mg\, y_p$$
$$V = -m\ell \sin\phi$$

The viscous friction of the cart and pendulum is represented using the Rayleigh dissipation function,

$$D = \frac{1}{2} b_c \dot{x}_c^2 + \frac{1}{2} b_p \dot{\phi}^2$$

The Lagrangian of the system is defined as

$$L = T^* - V$$

Substituting the co-kinetic and potential energy expressions gives

$$L = \frac{1}{2}(M+m)\,\dot{x}_c^2 + m\ell\dot{x}_c\dot{\phi}\sin\phi + \frac{1}{2}m\ell^2\dot{\phi}^2 + mg\ell\sin\phi$$

The Euler–Lagrange equation including viscous dissipation is

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{q}_i}\right) - \frac{\partial L}{\partial q_i} + \frac{\partial D}{\partial \dot{q}_i} = Q_i$$

For $q_1 = x_c$,

$$\frac{\partial L}{\partial \dot{x}_c} = (M+m)\,\dot{x}_c + m\ell\dot{\phi}\sin\phi$$

Taking the time derivative,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{x}_c}\right) = (M+m)\,\ddot{x}_c + m\ell\ddot{\phi}\sin\phi + m\ell\dot{\phi}^2\cos\phi$$

Also,

$$\frac{\partial L}{\partial x_c} = 0$$

and

$$\frac{\partial D}{\partial \dot{x}_c} = b_c\dot{x}_c$$

Since the generalised force associated with x_c is

$$Q_{x_c} = F$$

substitution into the Euler–Lagrange equation gives the cart equation

$$\boxed{F = (M+m)\,\ddot{x}_c + b_c\dot{x}_c + m\ell\ddot{\phi}\sin\phi + m\ell\dot{\phi}^2\cos\phi} \tag{1a}$$

Using the same Euler–Lagrange equation:

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) - \frac{\partial L}{\partial \phi} + \frac{\partial D}{\partial \dot{\phi}} = Q_\phi$$

For this system,

$$Q_\phi = 0$$

Considering the pendulum coordinate $q_2 = \phi$,

$$\frac{\partial L}{\partial \dot{\phi}} = m\ell \dot{x}_c \sin\phi + m\ell^2 \dot{\phi}$$

Taking the time derivative,

$$\frac{d}{dt}\left(\frac{\partial L}{\partial \dot{\phi}}\right) = m\ell \ddot{x}_c \sin\phi + m\ell \dot{x}_c \dot{\phi}\cos\phi + m\ell^2 \ddot{\phi}$$

Now differentiate L with respect to ϕ:

$$\frac{\partial L}{\partial \phi} = m\ell \dot{x}_c \dot{\phi}\cos\phi + mg\ell\cos\phi$$

For the dissipation term,

$$\frac{\partial D}{\partial \dot{\phi}} = b_p \dot{\phi}$$

Substituting into Euler–Lagrange:

$$m\ell \ddot{x}_c \sin\phi + m\ell \dot{x}_c \dot{\phi}\cos\phi + m\ell^2 \ddot{\phi} - \Big(m\ell \dot{x}_c \dot{\phi}\cos\phi + mg\ell\cos\phi\Big) + b_p\dot{\phi} = 0$$

The matching cross terms cancel, leaving

$$\boxed{m\ell^2\ddot{\phi} + b_p\dot{\phi} + m\ell\ddot{x}_c\sin\phi - mg\ell\cos\phi = 0} \tag{1b}$$

Q2: Analytical Linear Model of a Cart Pendulum System (40 marks)

The nonlinear model provided:

$$K_v \cdot u = (M+m)\,\ddot{x}_c + b_c\dot{x}_c + m\ell\ddot{\alpha}\cos\alpha - m\ell\dot{\alpha}^2\sin\alpha \tag{1a}$$

$$0 = m\ell^2\ddot{\alpha} + b_p\dot{\alpha} + m\ell\ddot{x}_c\cos\alpha + gm\ell\sin\alpha \tag{1b}$$

Q2.1

The system is linearised around $\alpha = 0$ corresponding to the pendulum vertically facing downwards.

For small angular displacements around $\alpha = 0$,

$$\sin\alpha \approx \alpha$$
$$\cos\alpha \approx 1$$

The nonlinear higher-order term is neglected:

$$\dot{\alpha}^2\sin\alpha \approx 0$$

Applying approximations to (1a):

$$K_v u = (M+m)\,\ddot{x}_c + b_c\dot{x}_c + m\ell\ddot{\alpha}(1) - m\ell\dot{\alpha}^2(0)$$

Applying approximations to (1b):

$$0 = m\ell^2\ddot{\alpha} + b_p\dot{\alpha} + m\ell\ddot{x}_c(1) + gm\ell(\alpha)$$

Therefore,

$$K_v u = (M+m)\,\ddot{x}_c + b_c\dot{x}_c + m\ell\ddot{\alpha} \tag{2a}$$

$$0 = m\ell^2\ddot{\alpha} + b_p\dot{\alpha} + m\ell\ddot{x}_c + gm\ell\alpha \tag{2b}$$

Q2.2

Assuming zero initial conditions, applying Laplace transform to (2a):

$$K_v U(s) = (M+m)\, s^2 X_c(s) + b_c s X_c(s) + m\ell s^2 A(s)$$

Taking Laplace transform of (2b):

$$0 = m\ell^2 s^2 A(s) + b_p s A(s) + m\ell s^2 X_c(s) + gm\ell A(s)$$

Therefore,

$$K_v U(s) = \left[(M+m)\, s^2 + b_c s\right] X_c(s) + m\ell s^2 A(s) \tag{3a}$$
$$0 = m\ell s^2 X(s) + \left[m\ell^2 s^2 + b_p s + gm\ell\right] A(s) \tag{3b}$$

$G_{x\alpha}(s)$:

Rearranging (3b),

$$\left[m\ell^2 s^2 + b_p s + gm\ell\right] A(s) = -m\ell s^2 X_c(s)$$

$$\frac{A(s)}{X_c(s)} = \frac{-m\ell s^2}{m\ell^2 s^2 + b_p s + gm\ell}$$

Therefore,

$$G_{x\alpha}(s) = \frac{A(s)}{X_c(s)} = -\frac{m\ell s^2}{m\ell^2 s^2 + b_p s + gm\ell} \tag{4a}$$
$$G_{x\alpha}(s) = -\frac{s^2}{\ell s^2 + \dfrac{b_p}{m\ell} s + g} \tag{4b}$$

$G_{ux}(s)$:

From (4a),

$$A(s) = -\frac{m\ell s^2}{m\ell^2 s^2 + b_p s + gm\ell} X_c(s)$$

Substituting into (3a),

$$K_v U(s) = \left[(M+m)\, s^2 + b_c s\right] X_c(s) + m\ell s^2 \left[-\frac{m\ell s^2}{m\ell^2 s^2 + b_p s + gm\ell} X_c(s)\right]$$

$$K_v U(s) = \left[(M+m)\, s^2 + b_c s - \frac{m^2\ell^2 s^4}{m\ell^2 s^2 + b_p s + gm\ell}\right] X_c(s)$$

Therefore,

$$\frac{X_c(s)}{U(s)} = \frac{K_v}{(M+m)\, s^2 + b_c s - \dfrac{m^2\ell^2 s^4}{m\ell^2 s^2 + b_p s + gm\ell}}$$

Multiplying the numerator and denominator by $m\ell^2 s^2 + b_p s + gm\ell$,

$$G_{ux}(s) = \frac{K_v\left(m\ell^2 s^2 + b_p s + gm\ell\right)}{\left[(M+m)\, s^2 + b_c s\right]\left[m\ell^2 s^2 + b_p s + gm\ell\right] - m^2\ell^2 s^4}$$

Expanding the denominator,

$$\begin{aligned}
&\left[(M+m)\,s^2+b_cs\right]\left[m\ell^2s^2+b_ps+gm\ell\right]-m^2\ell^2s^4\\
&=(M+m)\,m\ell^2s^4+(M+m)\,b_ps^3+(M+m)\,gm\ell s^2+b_cm\ell^2s^3+b_cb_ps^2+b_cgm\ell s-m^2\ell^2s^4\\
&=Mm\ell^2s^4+\left[(M+m)\,b_p+b_cm\ell^2\right]s^3+\left[(M+m)\,gm\ell+b_cb_p\right]s^2+b_cgm\ell s
\end{aligned}$$

Therefore,

$$G_{ux}(s)=\frac{K_v\left(m\ell^2s^2+b_ps+gm\ell\right)}{Mm\ell^2s^4+\left[(M+m)\,b_p+b_cm\ell^2\right]s^3+\left[(M+m)\,gm\ell+b_cb_p\right]s^2+b_cgm\ell s} \tag{5}$$

$G_{u\alpha}(s)$:

The overall cascaded transfer function is,

$$G_{u\alpha}(s)=G_{ux}(s)G_{x\alpha}(s)$$

Substituting (4a) and (5),

$$G_{u\alpha}(s)=\frac{K_v\left(m\ell^2s^2+b_ps+gm\ell\right)}{Mm\ell^2s^4+\left[(M+m)\,b_p+b_cm\ell^2\right]s^3+\left[(M+m)\,gm\ell+b_cb_p\right]s^2+b_cgm\ell s}\left[\frac{-m\ell s^2}{m\ell^2s^2+b_ps+gm\ell}\right]$$

Therefore,

$$G_{u\alpha}(s)=\frac{-K_vm\ell s^2}{Mm\ell^2s^4+\left[(M+m)\,b_p+b_cm\ell^2\right]s^3+\left[(M+m)\,gm\ell+b_cb_p\right]s^2+b_cgm\ell s} \tag{6}$$

Q2.3

From (1a), the effect of the pendulum angle over the cart is defined as,

$$d=\frac{m\ell}{K_v}\left(\ddot{\alpha}\cos\alpha-\dot{\alpha}^2\sin\alpha\right)$$

Thus,

$$u=\frac{1}{K_v}\left[(M+m)\,\ddot{x}_c+b_c\dot{x}_c\right]+d$$

For small values of m, ℓ and small angular oscillations,

$$d\approx 0$$

Therefore,

$$\begin{aligned}
u&=\frac{1}{K_v}\left[(M+m)\,\ddot{x}_c+b_c\dot{x}_c\right]\\
K_vu&=(M+m)\,\ddot{x}_c+b_c\dot{x}_c
\end{aligned}$$

Assuming zero initial conditions and applying the Laplace transform,

$$\begin{aligned}
K_vU(s)&=(M+m)\,s^2X_c(s)+b_csX_c(s)\\
&=\left[(M+m)\,s^2+b_cs\right]X_c(s)
\end{aligned}$$

Therefore,

$$\frac{X_c(s)}{U(s)} = \frac{K_v}{(M+m)\,s^2 + b_c s}$$

Hence,

$$\tilde{G}_{ux}(s) = \frac{K_v}{(M+m)\,s^2 + b_c s} \tag{7a}$$

$$\tilde{G}_{ux}(s) = \frac{K_v}{s\left[(M+m)\,s + b_c\right]} \tag{7b}$$

Q3: Data-Based System Model Identification (40 marks)

The known parameters of the laboratory setup are

$$M = 0.22\,\text{kg} \qquad m = 0.1\,\text{kg} \qquad g = 9.81\,\text{m/s}^2$$

while the friction coefficients b_c and b_p, the length to the centre of oscillation ℓ, and the input gain K_v are unknown and must be identified from the pulse-response data recorded in Lab 3.

Q3.1

Following Remark 2, the transfer function is identified between the input voltage $v_m(t)$ and the cart speed $v_c(t)$, and the integrator is appended afterwards. This avoids fitting the marginally stable pole at $s = 0$.

Since $v_c = \dot{x}_c$, that is $V_c(s) = sX_c(s)$, the simplified model (7b) gives

$$G_{uv_c}(s) = s\,\tilde{G}_{ux}(s) = \frac{K_v}{(M+m)\,s + b_c} \tag{8a}$$

$$G_{uv_c}(s) = \frac{a}{s+b} \qquad \text{where} \quad a = \frac{K_v}{M+m}, \quad b = \frac{b_c}{M+m} \tag{8b}$$

The structure in (8b) has one pole and no zeros. The MATLAB function `tfest()` is therefore called with `np = 1` and `nz = 0`, using the measured motor voltage `Vm` as input and the filtered derivative `vc` as output.

The identification returns

$$\hat{G}_{uv_c}(s) = \frac{0.887}{s + 16.25} \tag{9}$$

with a fit to the estimation data of 93.97 %.

Appending the integrator,

$$\tilde{G}_{ux}(s) = \frac{1}{s}\,\hat{G}_{uv_c}(s) = \frac{0.887}{s\,(s + 16.25)} \tag{10}$$

Q3.2

Following Remark 3, the transfer function is identified between the cart acceleration $a_c(t) = \ddot{x}_c(t)$ and the angle $\alpha(t)$, and the double derivative is applied afterwards.

Since $A_c(s) = s^2 X_c(s)$, substituting into (3b) gives

$$0 = m\ell\,A_c(s) + \left[m\ell^2 s^2 + b_p s + gm\ell\right] A(s)$$

Rearranging,

$$G_{a_c\alpha}(s) = \frac{A(s)}{A_c(s)} = \frac{-m\ell}{m\ell^2 s^2 + b_p s + gm\ell} \tag{11a}$$

$$G_{a_c\alpha}(s) = \frac{-\dfrac{1}{\ell}}{s^2 + \dfrac{b_p}{m\ell^2}s + \dfrac{g}{\ell}} \tag{11b}$$

The structure in (11b) has two poles and no zeros, so `tfest()` is called with `np = 2` and `nz = 0`, using the filtered cart acceleration `ac` as input and the measured angle `alpha_rad` as output.

The identification returns

$$\hat{G}_{a_c\alpha}(s) = \frac{-3.902}{s^2 + 0.5832s + 38.58} \tag{12}$$

with a fit to the estimation data of $78.4\,\%$.

Applying the double derivative,

$$\hat{G}_{x\alpha}(s) = s^2\,\hat{G}_{a_c\alpha}(s) = \frac{-3.902\,s^2}{s^2 + 0.5832s + 38.58} \tag{13}$$

Q3.3

Following Remark 4, the transfer function $G_{x\alpha}(s)$ is used to estimate ℓ and b_p, and the simplified transfer function $\tilde{G}_{ux}(s)$ is used to estimate K_v and b_c.

Writing the identified result (12) in the general second-order form

$$\hat{G}_{a_c\alpha}(s) = \frac{\hat{k}}{s^2 + \hat{a}_1 s + \hat{a}_0} \qquad \hat{k} = -3.902, \quad \hat{a}_1 = 0.5832, \quad \hat{a}_0 = 38.58 \tag{14}$$

and comparing coefficient by coefficient with the analytical expression (11b),

$$\hat{a}_0 = \frac{g}{\ell} \qquad \hat{a}_1 = \frac{b_p}{m\ell^2} \qquad \hat{k} = -\frac{1}{\ell} \tag{15}$$

From the first relation,

$$\ell = \frac{g}{\hat{a}_0} = \frac{9.81}{38.58}$$

$$\boxed{\ell = 0.2543\ \mathrm{m}} \tag{16}$$

Since $\ell = 2L/3$, the actual pendulum length is

$$L = \frac{3}{2}\,\ell = \frac{3}{2}\,(0.2543)$$

$$\boxed{L = 0.3814\ \mathrm{m}} \tag{17}$$

From the second relation,

$$b_p = m\ell^2\hat{a}_1 = (0.1)\,(0.2543)^2\,(0.5832)$$

$$\boxed{b_p = 3.771 \times 10^{-3}\ \mathrm{N\,m\,s/rad}} \tag{18}$$

The third relation in (15) is redundant and provides a consistency check on the identification:

$$-\frac{1}{\hat{k}} = \frac{1}{3.902} = 0.2563\ \mathrm{m} \tag{19}$$

which agrees with (16) to within 0.79 %.

Turning to the cart, the identified result (9) is written in the general first-order form

$$\hat{G}_{uv_c}(s) = \frac{\hat{a}}{s + \hat{b}} \qquad \hat{a} = 0.887, \quad \hat{b} = 16.25 \tag{20}$$

Comparing with the analytical expression (8b), $\hat{a} = K_v/(M+m)$ and $\hat{b} = b_c/(M+m)$, so with $M + m = 0.32$ kg,

$$K_v = (M+m)\,\hat{a} = (0.32)\,(0.887)$$
$$\boxed{K_v = 0.2838\,\mathrm{N/V}} \tag{21}$$

and

$$b_c = (M+m)\,\hat{b} = (0.32)\,(16.25)$$
$$\boxed{b_c = 5.200\,\mathrm{N\,s/m}} \tag{22}$$

The complete set of identified parameters is summarised below.

Parameter	Expression	Value	Units
ℓ	$g/\hat{a}_0$	0.2543	m
L	$3\ell/2$	0.3814	m
b_p	$m\ell^2\hat{a}_1$	3.771×10^{-3}	N m s/rad
K_v	$(M+m)\,\hat{a}$	0.2838	N/V
b_c	$(M+m)\,\hat{b}$	5.200	N s/m

These values are physically consistent with the recorded response. From (11b) the pendulum natural frequency, oscillation period and damping ratio are

$$\omega_n = \sqrt{\hat{a}_0} = 6.211\ \mathrm{rad/s} \qquad T = \frac{2\pi}{\omega_n} = 1.012\ \mathrm{s} \qquad \zeta = \frac{\hat{a}_1}{2\sqrt{\hat{a}_0}} = 0.0469 \tag{23}$$

matching the lightly damped, approximately one-second oscillation observed in the experimental data. For the cart, the time constant and steady-state speed gain are

$$\tau = \frac{M+m}{b_c} = 61.5\ \mathrm{ms} \qquad \frac{K_v}{b_c} = 0.0546\ (\mathrm{m/s})/\mathrm{V} \tag{24}$$

Q3.4

Nonlinear model. Rearranging (1a) for $\ddot{x}_c$ and (1b) for $\ddot{\alpha}$ gives the two expressions implemented in Simulink:

$$\ddot{x}_c = \frac{K_v u - b_c\dot{x}_c - m\ell\ddot{\alpha}\cos\alpha + m\ell\dot{\alpha}^2\sin\alpha}{M+m} \tag{25}$$

$$\ddot{\alpha} = -\frac{b_p}{m\ell^2}\dot{\alpha} - \frac{1}{\ell}\ddot{x}_c\cos\alpha - \frac{g}{\ell}\sin\alpha \tag{26}$$

Each expression is built as a separate subsystem, with the accelerations fed through the integrator chains $\ddot{x}_c \to \dot{x}_c \to x_c$ and $\ddot{\alpha} \to \dot{\alpha} \to \alpha$, all initialised at zero. No small-angle approximation is applied, so the $\sin\alpha$, $\cos\alpha$ and $\dot{\alpha}^2$ terms are retained in full.

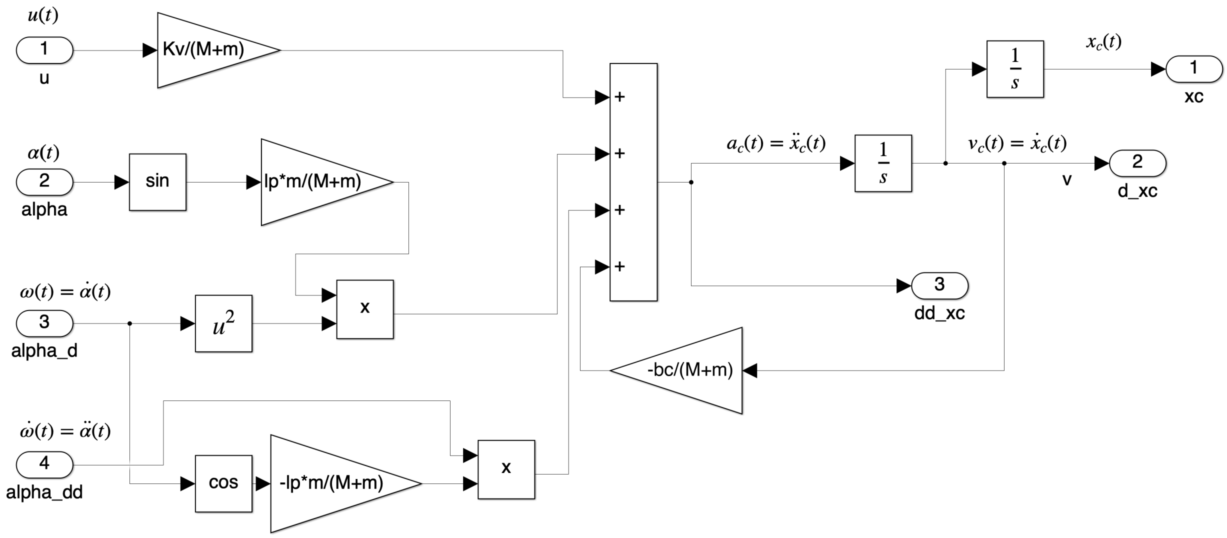

Figure 1: Nonlinear subsystem implementing (25), the cart acceleration $\ddot{x}_c$.

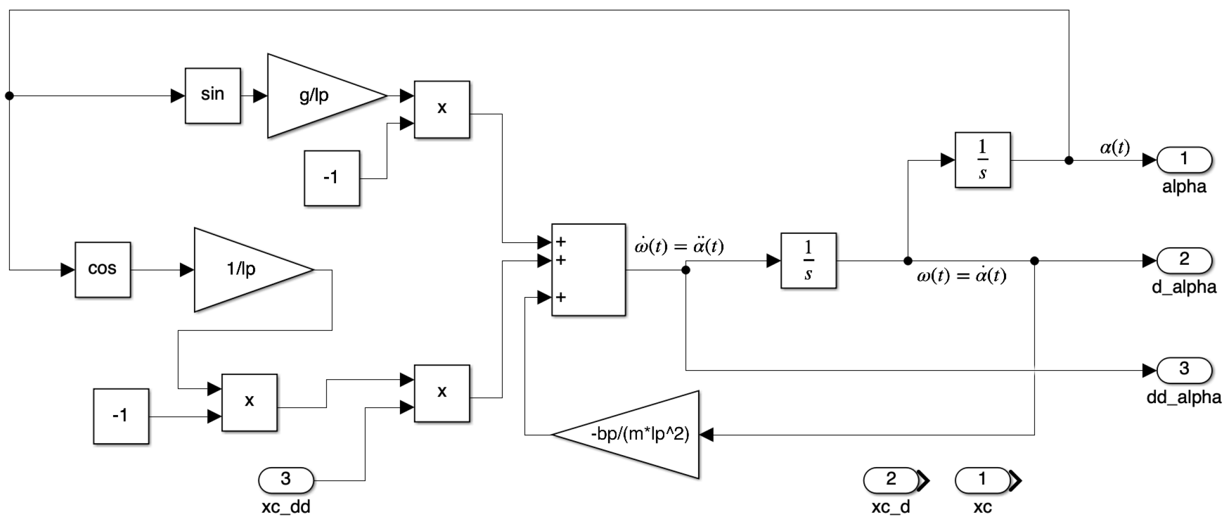

Figure 2: Nonlinear subsystem implementing (26), the angular acceleration $\ddot{\alpha}$.

Because (25) requires $\ddot{\alpha}$ and (26) requires $\ddot{x}_c$, the two subsystems form an algebraic loop. The loop is well posed: substituting (26) into (25) shows that the loop gain is

$$\frac{m\cos^2\alpha}{M+m} \leq \frac{0.1}{0.32} = 0.3125 < 1 \tag{27}$$

so the solver converges for every value of α.

Linear model. As required by Remark 6, the linear branch uses the full transfer function $G_{ux}(s)$ of (5) — not the simplified $\tilde{G}_{ux}(s)$ — cascaded with $G_{x\alpha}(s)$ of (4a). Substituting the identified parameters into (5) and (4a),

$$G_{ux}(s) = \frac{0.0018352\,s^2 + 0.0010703\,s + 0.070803}{0.0014224\,s^4 + 0.034828\,s^3 + 0.099431\,s^2 + 1.2971\,s} \tag{28}$$

$$G_{x\alpha}(s) = \frac{-0.025428\,s^2}{0.0064657\,s^2 + 0.0037708\,s + 0.24945} \tag{29}$$

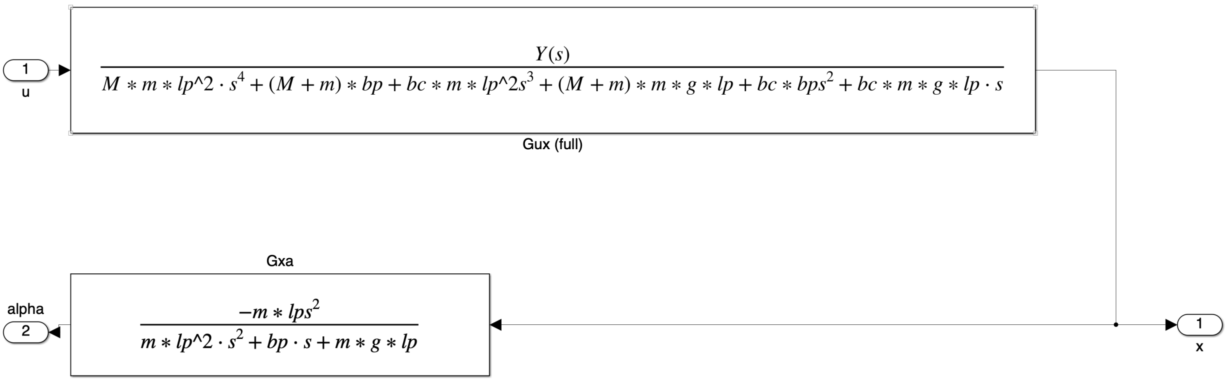

Figure 3: Linear branch: the full $G_{ux}(s)$ of (28) cascaded with $G_{x\alpha}(s)$ of (29).

Comparison. Both branches are driven by the same input, a 4 V pulse of one second duration applied at $t = 2$ s.

During the pulse the cart accelerates and settles to the steady-state speed predicted by (24), travelling approximately 0.217 m before coming to rest when the voltage is removed. Because $G_{ux}(s)$ contains a free integrator and there is no position feedback, the cart holds that displacement for the remainder of the simulation. The pendulum is driven to a peak deflection of about $-6.8°$ at the onset of the pulse and then rings down with the period and damping ratio predicted by (23).

The two models are visually indistinguishable across the whole run. The mismatch remains below 0.2 mm in x_c and below $0.1°$ in α, that is under 0.1 % of the respective signal amplitudes.

This is the expected outcome and it quantifies the validity of the linearisation assumptions made in Q2.1. At the observed peak of $|\alpha| = 6.8° = 0.1187$ rad, the approximation $\sin\alpha \approx \alpha$ carries a relative error of only 0.235 % and $\cos\alpha \approx 1$ an error of 0.703 %. The neglected term $m\ell\dot{\alpha}^2\sin\alpha$ peaks at approximately 1.64 mN, which is 1.42 % of the corresponding retained term $m\ell\ddot{\alpha}\cos\alpha$ and only 0.144 % of the applied force $K_v u = 1.135$ N.

The two models would only be expected to separate visibly for a substantially larger excitation, where $|\alpha|$ grows beyond roughly $30°$ and both the curvature of $\sin\alpha$ and the quadratic velocity term become significant.

References

[1] R. P. Aguilera, *Lecture Notes on System Modelling*, University of Technology Sydney, Australia, 2026.

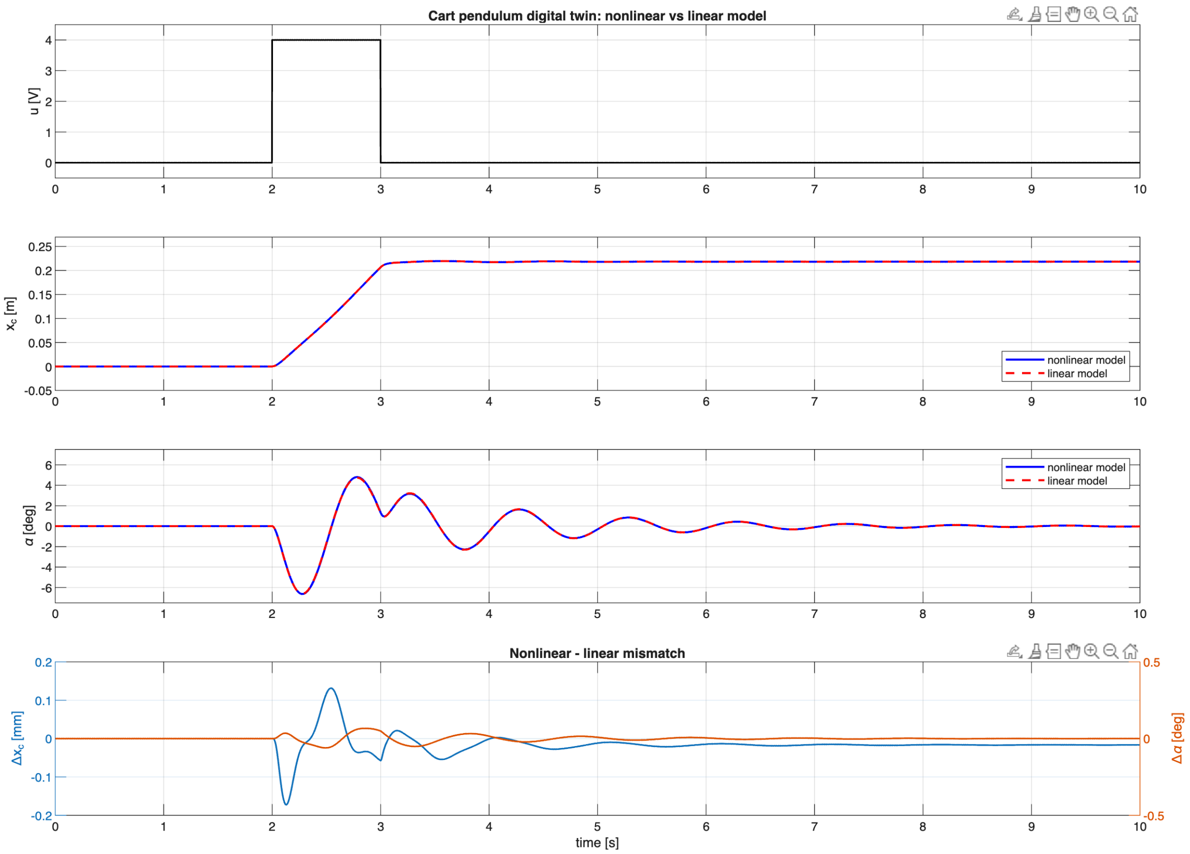

Figure 4: Simulated response of the nonlinear and linear models to the same input: applied voltage, cart position, pendulum angle, and the mismatch between the two models.

[2] B. Messner and D. Tilbury, *Control Tutorials for MATLAB and Simulink*, University of Michigan, USA. [Online]. Available: `http://ctms.engin.umich.edu/CTMS`

[3] L. Ljung, *System Identification Toolbox User's Guide*, The MathWorks. [Online]. Available: `https://ch.mathworks.com/help/pdf_doc/ident/ident_ug.pdf`